

MIT 18.06 - 3.1

Problem 24.1

a) Solve $\det \begin{pmatrix} 1-\lambda & b \\ b & 1-\lambda \end{pmatrix} = 0 \therefore$

$$(1-\lambda)^2 - b^2 = 0$$
$$\lambda^2 - 2\lambda + 1 - b^2 = 0$$

$$\lambda = \frac{2 \pm \sqrt{4 - 4(1-b^2)}}{2} < 0$$

$$2 + \sqrt{4 - 4(1-b^2)} < 0$$

$$4 < 4 - 4(1-b^2)$$

$$0 < -4 + 4b^2$$

$$1 < b^2$$

So eigenvalues are negative $\forall b$ such that $b < -1$ or $b > 1$

b) ~~The~~ $\begin{bmatrix} 1 & b \\ b & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & b \\ 0 & 1-b^2 \end{bmatrix}$ a negative

Since we showed earlier in part a that $\uparrow$ eigenvalue $\&$ existing requires $b^2 > 1$, $1-b^2 < 0$ i.e. negative.

c) eig. There are a total of 2 eigenvalues and they must add to the

they must add to the trace i.e.
2. But two negative numbers cannot have a positive sum, so at least one eigenvalue must be positive.

Problem 24.2

A is invertible because $\det(A) = -1$.

" " orthogonal " $AA^T = I$

" " not projection " $A^2 \neq A$.

" " permutation " each column has a 1 and zeroes everywhere else

" " diagonalizable " A is symmetric.

" " Markov because columns sum to one and all entries in range $[0, 1]$

LU is not possible, ~~no~~ because ~~no~~ pivot in first column 1,1 without row exchange position
QR possible because A is orthonormal
has full column rank, SAS^{-1} possible because A is diagonalizable, $Q\Lambda Q^{-1}$ is possible because A is symmetric.

B is not invertible because $\det(B) = 0$

" " " orthogonal " $BB^T \neq I$

" " " projection " $B = B^T$ and $B^2 = B$

" " " permutation " every column does not have a single non zero entry equal to 1.

B is diagonalizable because B is symmetric.

B is markov because every entry in the range $[0, 1]$ and every column adds to 1.

LU is possible but U may will have two zeroes on the diagonal.

QR is ~~not~~ not possible because columns are dependent. $S \Lambda S^{-1}$ possible because B is diagonalizable.

$Q \Lambda Q^T$ possible because B is symmetric.

Problem 24.3

$$A = \begin{bmatrix} 0.7 & 0.1 & 0.2 \\ 0.1 & 0.6 & 0.3 \\ 0.2 & 0.3 & 0.5 \end{bmatrix}$$

Steady state vector $\vec{x}$ is $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ because $Ax = x$ i.e. it is the eigenvector with eigenvalue 1.

When A is symmetric, $A = A^T$. Thus the rows of A sum to 1, ~~so~~ $Ax = x$ for $x = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ so the rows of $A - I$ sum to 0.

The steady state vector is the eigenvector with eigenvalue 1. So $(A - I)x = 0$ gives $x = (1, \dots, 1)$.